

REUBEN SAMSON RAJ

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EDUCATION

Ph.D. in Computer Science , University of Arkansas, Fayetteville, AR Advisor: Dr. Kevin Jin Dissertation: <i>Frameworks on Network Programmability and Abstraction for Securing and Strengthening Smart-Grids</i>	Aug 2021 – Dec 2026 (expected)
M.S. in Computer Engineering , University of Arkansas, Fayetteville, AR	Aug 2021 – Dec 2024
B.E. in ECE , College of Engineering, Guindy, Anna University, Chennai, India	Jun 2006 – May 2010

RESEARCH INTERESTS

Network Security, Programmable Networks, Smart-Grid Networks, Cloud-Native AI

WORK EXPERIENCE

Graduate Research Assistant <i>University of Arkansas</i>	Aug 2021 – Present Fayetteville, AR
Network Architecture Research Intern <i>Nokia Bell Labs</i>	Jun 2023 – Aug 2023 Murray Hill, NJ
Software Engineer <i>Cisco Systems</i>	Jun 2020 – Jul 2021 Bangalore, India
Member of Technical Staff <i>Parallel Wireless</i>	Oct 2018 – May 2020 Bangalore, India
R&D Engineer <i>Nokia Networks</i>	Apr 2015 – Oct 2018 Bangalore, India
Software Engineer <i>Cisco Systems</i>	Apr 2012 – Apr 2015 Bangalore, India
Associate Software Engineer <i>Accenture</i>	Apr 2011 – Mar 2012 Chennai, India

PUBLICATIONS

- Reuben Samson Raj**, Dong Jin. *A Framework to Evaluate PMU Networks for Resiliency Under Network Failure Conditions*. IEEE SmartGridComm 2022 (📄)
- Zhiyao He, Yanfeng Qu, Gong Chen, **Reuben Samson Raj**, Hui Lin, Dong Jin. *Towards Secure and Resilient Synchronphasor Networks Using P4 Programmable Switches*. IEEE GreenTech 2024 (📄)
- Reuben Samson Raj**, Dong Jin. *Dynamic Data Driven Security Framework for Industrial Control Networks using Programmable Switches*. DDDAS 2024 (📄)
- Reuben Samson Raj**, Dong Jin. *Leveraging Data Plane Programmability Towards a Policy-driven In-Network Security Framework for Industrial Control Systems*. ACM e-Energy 2025 (📄)
- Reuben Samson Raj**, Dong Jin. *A Dynamic Data-Driven Framework for Enhancing Energy System Security and Resilience through Programmable Networks*. Book Chapter - DDDAS Handbook Vol. IV, Springer (Under Production). (📄)

6. Sarvesh Bidkar, Rakesh Abbireddy, Maryam Amiri, **Reuben Samson Raj**, Anil Rana, Jeff McLaird, Paul Rea, Meryem Simsek, Jesse E. Simsarian. *Demonstration of Multi-Provider Network and Cloud Service Provisioning with Blockchain Smart Contracts*. ECOC 2025 (📄)
7. Zhiyao He*, **Reuben Samson Raj***, Yanfeng Qu, Dong Jin. *Programmable Data Plane Approaches for ML-Based Anomalous Event Classification in Grid Sensor Networks*. IEEE ICC 2026 (Accepted, * - Equal Contributions) (📄)
8. Yuexin Chen, **Reuben Samson Raj**, Xiaoyu Guo, Dong Jin. *Enhancing Security of Grid Control Systems through In-network Collaboration of Programmable Data Planes*. IEEE SmartGridComm 2026 (Accepted)
9. Haden Fowler, **Reuben Samson Raj**, Dong Jin. *Dynamic Data-Driven Cyber Defense for Industrial Control Systems using P4-Based Cross-Layer Telemetry*. DDDAS 2026 (Under Review)

AWARDS & HONORS

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|---|------|
| → 2nd place - Best Paper Award, DDDAS Conference | 2024 |
| → Student Travel Grant, ACM e-Energy | 2025 |
| → Larry & Gwen Stephens Graduate Scholarship | 2024 |
| → Reginald R. Barney & Jameson A. Baxter Graduate Scholarship | 2023 |
| → Department of Electrical Engineering and Computer Science Scholarship | 2022 |
| → You Amaze and You Inspire Awards, Cisco | 2020 |
| → Best Feature Team Award, Nokia | 2016 |
| → CAP Award, Cisco | 2013 |

ACADEMIC SERVICE & VOLUNTEERING

- Reviewer, *ACM Transactions on Modeling and Computer Simulation (TOMACS)*, 2025, 2026
- Reviewer, *IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, 2026
- Mentoring and guidance for undergraduate honor's thesis projects
- Served as vendor liaison and key coding platform point-of-contact for regional high-school programming contests

SKILLS

Languages	Python, P4 (Tofino, bmv2), C, Bash
Networking	SDN, P4, 5G Core, EPC, TCP/IP, Mininet, Wireshark, iperf
Cloud & Orchestration	Kubernetes, Docker, OpenStack
Security Tools	ICS/OT (Modbus, DNP3), Network Security, IDA Pro, PE Studio
Tools & Frameworks	Linux, Git, FastAPI, OpenCV, NVIDIA GPU Stack